

Static labor wedge

efficiency: $mpl(l_t) = mrs(G_t, l_t) = \frac{U_l(G_t, l_t)}{U_c(G_t, l_t)}$ leisure $l_t = 1 - n_t$
l: countercyclical - C procyclical $\Rightarrow$ mrs: procyclical
avg. productivity or mpl is not procyclical

Chari, Kehoe, McGrattan (2007): Business cycle accounting (Labor wedge)

Wedge: $\ln(\frac{mpl}{mrs})$ Data: countercyclical
 $= \text{const.} + \ln(\frac{Y_t}{N_t}) - \frac{1}{\eta} \ln N_t - \frac{1}{\sigma} \ln G_t$

Why? ① No RA Booms: lower-wage worker enter
Consumption drops if unemployed
search frictions

② Mismatched mpl or mrs mpl is more procyc. CES
mrs is more countercyc. GHH (?)

③ sticky wages: mute cyclical of MRS

Karabarbounis 2014: Source of labor wedge

Decomposition: $\text{wedge} = \ln(\frac{mpl}{mrs}) = \ln(\frac{mpl}{w/p}) + \ln(\frac{w/p}{mrs})$
Sticky price, markups $\leftarrow$ product-market $\leftarrow$ labor-market
Measurement: $\frac{mpl}{w/p} = \frac{P}{W} \cdot \frac{Y}{N} = \frac{1}{\text{labor share } (\chi)}$ e.g. sticky wage
Data: AHE: average hourly earnings
Problem: Spot market (Becker 1964, Hall 1980)
Alternatives: measure product-market wedge (Bils, Klenow, Malin 2018)
user cost of labor } Kudlyak 2014
Bils, Kudlyak, Lims 2023

Boppart, Krusell (2020): long-run labor supply $\downarrow$ WE > SE

BK preference: $U(C, h) = \frac{1}{1-\sigma} \{ [C \cdot v(h c^{\frac{1}{\sigma}})]^{1-\sigma} - 1 \}$ $\sigma \neq 1$
 $\ln C + \ln v(h c^{\frac{1}{\sigma}})$ $\sigma = 1$
 $mrs = \frac{v'(h) \cdot h \cdot \frac{1}{c}}{v(h) + v'(h) \cdot h \cdot \frac{1}{c}} = w \Rightarrow \chi = h c^{\frac{1}{\sigma}} = \text{const.}$

$\Rightarrow \chi_h = -\frac{1}{\sigma} \chi_c$ $v \approx 0.2$
McGrudly 1981: $U(C, h) = \frac{C^{1-\sigma} - 1}{1-\sigma} - \varphi \frac{h^{1+\frac{1}{\sigma}}}{1+\frac{1}{\sigma}}$ $\sigma > 1$
 $MRS = \frac{\varphi h^{\frac{1}{\sigma}}}{C} = w \Rightarrow \frac{\varphi h^{\frac{1}{\sigma}}}{C} = \frac{w}{C} = \text{const.}$

Generalized log-log: $U(C, h) = \ln(C) + k \ln(1 - \phi h c^{\frac{1}{\sigma}})$ $v(\phi, \sigma)$
GHH: $U(C, h) = \frac{1}{1-\sigma} \{ C - \varphi h^{\frac{1}{\sigma}} / (1-\sigma) \}^{1-\sigma}$ $\sigma \neq 1$

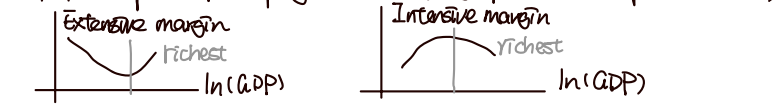
Comments: US post-WWII: hours - ?

① Home production: total work $\downarrow$ = market + home $\downarrow$
Aguiar & Hurst 2007: Leisure $\uparrow$ (Age $\uparrow$ home prod. $\uparrow$)

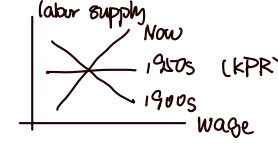
② Nature of work evolves
Epstein, Kimball 2021: Job utility $\uparrow$

③ New/ Better leisure
Aguiar, Bils, Charles, Hurst (2021): Recreational computer $\uparrow$

Bick, Fuchs-Schündeln, Lagakos (2018): Cross-country hours worked
GDP per capita $\uparrow$ employment $\downarrow$ hours per adult $\downarrow$



Labor Supply Curve in US. ① Taxes & redistribution



② Preference.

Epstein, Kimball 2021
Ferraro, Valaitis 2025: Consump. quality

Richard Rogerson 1988 JME: Indivisible labor & lotteries

$\max_n n U(C, 1-h-\phi) + (1-n) U(C, 1)$ st. $n \cdot h = X$ total labor
 $\Rightarrow h^* U_n(C, 1-h^*-\phi) = U(C, 1) - U(C, 1-h-\phi)$
 h^* does not depend on X : "constant margin"
 $\Rightarrow U(C, n) = U(C, 1) - n \cdot [U(C, 1) - U(C, 1-h-\phi)] \text{ Const.}$

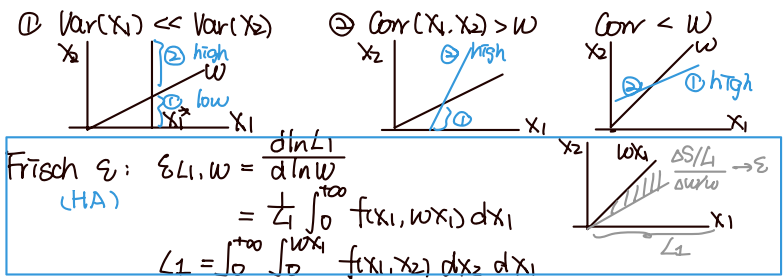
Total utility is linear in extensive margin n . Frisch $\epsilon = +\infty$

Roy model 1964: Self selection. (HA)

Utility = $\max \{ w_1 x_1, w_2 x_2 \}$
Work in sector 2 $\Leftrightarrow x_2 \geq \frac{w_1}{w_2} x_1 =: w x_1$

Heckman selection bias: E[income sectors] on truncated distribution

key to have high & low income sectors:



Chang, Kim

Work in market $\Leftrightarrow U(C, 1-h) + w x_1 h \geq U(C, 1)$
 $\Leftrightarrow w x_1 \geq \phi$

Chang, Kim (2006/2007): Heterogeneous agent labor supply

Fact: Micro Frisch $\epsilon \approx 0.5$, Macro Frisch $\epsilon \approx 2-4$ (RBC)
Model: Indivisible labor, HA (Aiyagari). 2-earner family
Reservation wage w^R (a: asset, x: productivity):
Frisch $\epsilon \propto pdf(w^R) \approx 1.0$

Problem: Ex-ante HA: Too high transition between E & U.
No long-run labor supply diff.
Data: wage $\uparrow$ hours $\uparrow$

Mustre-Del-Rio, 2015: Ex-ante heterogeneity: less ϵ than CK

Fact: Chang & Kim: wealth $\uparrow$ employment $\downarrow$. Not true in data.
Model: Disutility on labor ϕ_i , Market skill x_i .

$\phi_i \downarrow x_i \uparrow \Rightarrow$ employment $\uparrow \Rightarrow$ wealth $\uparrow$ "Composition effect"
Park 2020: Consumption.
 $E(C) = \frac{g_{w/c}(w^R(c))}{1 - g_{w/c}(w^R(c))} \rightarrow 0.4$
Quantile / Trimming method
asset: pref. market friction

Gimpelson 2024: Retirement hazard on work history

Retrospective data: match SOEP with SOEP-RU
Friction matters } On & off labor supply: Majority chooses max (emp)
Concordance rate $\uparrow$ without friction

The Labor Wedge & Labor Demand

Bils, Klenow, Molin (2018): Product Market Wedge (intermediates)

Intensive & extensive margin: μ^P should affect them simultaneously.

AHE can't represent true "Price of labor":

- ① Quality: Recession: "low skill, low wage" out. AHE -
- ② Smoothing: Implicit contract Hall 1980: paycheck $\neq$ spot price

New-hire wage? Quality bias: Recession: hire $\uparrow$, wage $\uparrow$

User cost of labor Kudiyak 2014 Pro-cyclical!

$$UC_t = W_{t,t} + \sum_{s=1}^{\infty} \beta^s (1-\delta)^s (W_{t,t+s} - W_{t+1,t+s})$$

Alternative decomposition:

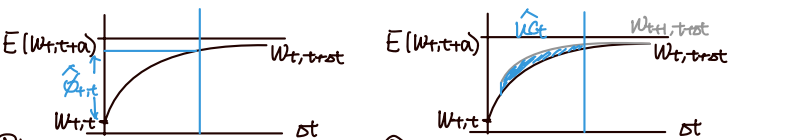
- ① Self-employed: $\mu = \mu^P = \frac{P}{mc}$ ← Both countercyclical
- ② Intermediate inputs: $\mu^P = \frac{P}{mc} = \frac{P \cdot mpm}{P_m} = \frac{P}{P_m} \left(\frac{y}{m} \right)^{\frac{1}{\sigma}-1}$ ↓

Bils, Kudiyak, Lins (2023): Quality-adjusted

- QA UC procyclical: New-hire wage: procyc
- Lock-in effect (smoothing): procyc
- Separation effect (quality): countercyc.

$$Cov(Cyc_t, \ln \hat{q}_{t,t}) = Cov(Cyc_t, \ln W_{t,t} - \ln W_{t,t+a})$$

$$Cov(Cyc_t, \ln UC_t^Q) = Cov(Cyc_t, \ln W_{t,t} - \ln W_{t,t+a} + \sum_{s=1}^a \Delta_{t,t+s} [(\ln W_{t,t+s} - \ln W_{t,t+a}) - (\ln W_{t+1,t+s} - \ln W_{t+1,t+a})])$$



- ① If wage smoothing persists: $Cov(Cyc_t, W_{t,t+a}) > 0$
- ② If quality growth is higher for match: In recession: $Cov(Cyc_t, \ln \frac{q_{t,t+a}}{q_{t,t}}) < 0$

Understate procyc. of New-Hire wages Overstate procyc. of UC & NH

- ③ Imputation: impute wage growth from survived matches. $\Rightarrow$ Overstate procyc.

Cyc. of PMW: New Keynesian: sticky price: $Cyc \downarrow, MC \downarrow, Markup \uparrow, L^D \downarrow$

Financial friction: liquid constraint: Markup $\uparrow, L^D \downarrow$

Gilchrist, Schoenle, Sim, Zakrajsek (2017)

Customer market vs liquidity constraint

Arellano, Bai, Kehoe (2019)

Fact: Recession. Cross-sectional dispersion of firm growth $\uparrow$ mpl -

Risky input hiring. FF: incomplete market, default cost, agency cost

Volatility $\sigma \uparrow$ scale $\downarrow$ wedge $= \ln(mpl) - \ln(wages) \uparrow$

$$\max_{\hat{z}} \int_{\hat{z}}^{\infty} (z^{\theta} - w_l + v) f(z) dz, \text{ s.t. } \hat{z}^{\theta} - w_l = 0$$

$$\theta \hat{z}^{\theta-1} E\{z | z \geq \hat{z}\} = w + v \frac{f(\hat{z})}{1-F(\hat{z})} \frac{d\hat{z}}{dw} \text{ wedge}$$

Kehoe, Lopez, Midrigan, Pastorino (2020): Habit & learning-by-doing

$$\frac{1}{1-\alpha} E_0 \sum_{t=0}^{\infty} \beta^t (C_t - x_t)^{1-\alpha} \cdot MU_t \propto (C_t - x_t)^{-\alpha} = C_t^{-\alpha} \left(\frac{C_t - x_t}{C_t} \right)^{-\alpha}$$

Recession: $G_t \approx X_t, C_t \downarrow, MU_t \uparrow$ MRS = stochastic discount factor $\uparrow$

PV of match $\downarrow$ vacancy $\downarrow$ labor wedge $\uparrow$

Human capital accumulation: Matching as a long-duration asset.

AHH preference: No wealth effect ($V' > 0$)

$$U(C, n) = \frac{1}{1-\sigma} \left(C - \frac{n^{\mu}}{H^{\nu}} \right)^{1-\sigma} \text{ s.t. } C = wn + T$$

$$FOC: \left(C - \frac{n^{\mu}}{H^{\nu}} \right)^{-\sigma} = \pi \Rightarrow n^{\nu} = w$$

$$\left(\right)^{-\sigma} n^{\nu} = w \pi$$

$$MRS = \left| \frac{U_n}{U_c} \right| = n^{\nu}$$

KPR preference: Balanced wealth & consumption effect.

$$U(C, n) = \ln C + \ln(v(1+n)) \text{ or } \frac{1}{1-\sigma} \left(C \cdot v(1+n) \right)^{1-\sigma}$$

$$MRS = - \frac{U_n}{U_c} = C \cdot \frac{v'(1+n)}{v(1+n)} = w \Rightarrow \frac{v'(1+n)}{v(1+n)} = \frac{w}{C} \text{ BLP. } n = n^*$$

(Kaldor facts)